Hongjie Fang

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Education

Shanghai Jiao Tong University *2022/09 – 2027/06 (expected)*
Ph.D. Candidate, Wu Wenjun Honorable Class, Computer Science Shanghai, China

Shanghai Jiao Tong University *2018/09 – 2022/06*
Bachelor of Engineering, major in Computer Science and Engineering Shanghai, China
Bachelor of Economics, minor in Finance

- GPA 4.03 / 4.3, Ranking: 2 / 149.

Awards

Best Paper Award, ICRA 2024 *2024*
Best Paper Award, IROS 2023 Manipulation and Grasping Workshop *2023*
Shanghai Outstanding Graduates *2022*
Shanghai Scholarship *2020*
National Olympics in Informatics (**NOI**) *Bronze Medal* *2017*
National Olympics in Informatics in Provinces (**NOIP**) *First Prize* *2015 & 2016*

Publications & Preprints **Total Citations: 4115**

* denotes equal contribution, # denotes corresponding author(s).

1. Force Policy: Learning Hybrid Force-Position Control Policy under Interaction Frame for Contact-Rich Manipulation
   Hongjie Fang*, Shirun Tang*, Mingyu Mei*, Haoxiang Qin, Zihao He, Jingjing Chen, Ying Feng, Chenxi Wang, Wanxi Liu, Zaixing He, Cewu Lu, Shiquan Wang#
   *Robotics: Science and Systems (**RSS**), 2026.*

2. AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons
   Hongjie Fang*, Chenxi Wang*, Yiming Wang*, Jingjing Chen*, Shangning Xia, Jun Lv, Zihao He, Xiyan Yi, Yunhan Guo, Xinyu Zhan, Lixin Yang, Weiming Wang, Cewu Lu#, Hao-Shu Fang#
   *Conference on Robot Learning (**CoRL**), 2025, **(Oral Presentation)**.*

3. AirExo: Low-Cost Exoskeletons for Learning Whole-Arm Manipulation in the Wild
   Hongjie Fang*, Hao-Shu Fang*, Yiming Wang*, Jieji Ren, Jingjing Chen, Ruo Zhang, Weiming Wang, Cewu Lu#
   *IEEE International Conference on Robotics and Automation (**ICRA**), 2024.*

4. TransCG: A Large-Scale Real-World Dataset for Transparent Object Depth Completion and A Grasping Baseline
   Hongjie Fang, Hao-Shu Fang, Sheng Xu, Cewu Lu#
   *IEEE Robotics and Automation Letters (**RA-L**), 2022; Presented at **ICRA** 2023.*

5. History-Aware Visuomotor Policy Learning via Point Tracking
   Jingjing Chen*, **Hongjie Fang***, Chenxi Wang, Shiquan Wang#, Cewu Lu#
   *IEEE International Conference on Robotics and Automation (**ICRA**), 2026.*

6. FoAR: Force-Aware Reactive Policy for Contact-Rich Robotic Manipulation
   Zihao He*, **Hongjie Fang***, Jingjing Chen, Hao-Shu Fang#, Cewu Lu#
   *IEEE Robotics and Automation Letters (**RA-L**), 2025; Presented at **IROS** 2025.*

7. Learning Dexterous Manipulation with Quantized Hand State
   Ying Feng*, **Hongjie Fang*#**, Yinong He*, Jingjing Chen, Chenxi Wang, Zihao He, Ruonan Liu, Cewu Lu#
   *IEEE International Conference on Robotics and Automation (**ICRA**), 2026.*

8. Asynchronous Multimodal Diffusion Policy Composition via Latency-Aware Guidance Fusion
   Zihao He*, **Hongjie Fang***, Shirun Tang, Cewu Lu#, Hao-Shu Fang#
   RSS 2026 Workshop on Diffusion for Robot Learning, 2026.

9. AnyDexRT: Calibration-Free Dexterous Hand Retargeting with Few-Shot Human Guidance
Chenxi Wang*, Ying Feng*, **Hongjie Fang#**, Shangning Xia, Lixin Yang, Chuan Wen, Cewu Lu#
RSS 2026 Workshop on Dexterous Manipulation, 2026.
10. RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot
Hao-Shu Fang, **Hongjie Fang**, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, Haoyi Zhu, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2024*
11. RISE: 3D Perception Makes Real-World Robot Imitation Simple and Effective
Chenxi Wang, **Hongjie Fang**, Hao-Shu Fang#, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024.*
12. CAGE: Causal Attention Enables Data-Efficient Generalizable Robotic Manipulation
Shangning Xia, **Hongjie Fang**, Cewu Lu#, Hao-Shu Fang#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2025.*
13. Towards Effective Utilization of Mixed-Quality Demonstrations in Robotic Manipulation via Segment-Level Selection and Optimization
Jingjing Chen, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2025.*
14. X-Imitator: Spatial-Aware Imitation Learning via Bidirectional Action-Pose Interaction
Kai Xiong, **Hongjie Fang**, Lixin Yang, Cewu Lu#
arXiv preprint, 2026.
15. AnyGrasp: Robust and Efficient Grasp Perception in Spatial and Temporal Domains
Hao-Shu Fang, Chenxi Wang, **Hongjie Fang**, Minghao Gou, Jirong Liu, Hengxu Yan, Wenhai Liu, Yichen Xie, Cewu Lu#
*IEEE Transaction on Robotics (**T-RO**), 2023; Presented at **ICRA** 2024, **(Best Paper Award on IROS 2023 Workshop)**.*
16. Motion Before Action: Diffusing Object Motion as Manipulation Condition
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang#
*IEEE Robotics and Automation Letters (**RA-L**), 2025; Presented at **ICRA** 2026.*
17. Dense Policy: Bidirectional Autoregressive Learning of Actions
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Han Xue, Hao-Shu Fang, Yong-Lu Li, Cewu Lu, Lixin Yang#
*IEEE/CVF International Conference on Computer Vision (**ICCV**), 2025.*
18. Knowledge-Driven Imitation Learning: Enabling Generalization Across Diverse Conditions
Zhuochen Miao*, Jun Lv*, **Hongjie Fang**, Yang Jin, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.*
19. Open X-Embodiment: Robotic Learning Datasets and RT-X Models
Open X-Embodiment Collaboration, 147 authors
*IEEE International Conference on Robotics and Automation (**ICRA**), 2024, **(Best Paper Award)**.*
20. Flexible Handover with Real-Time Robust Dynamic Grasp Trajectory Generation
Gu Zhang, Hao-Shu Fang, **Hongjie Fang**, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2023.*
21. AnyDexGrasp: Learning General Dexterous Grasping for Any Hands with Human-level Learning Efficiency
Hao-Shu Fang#, Hengxu Yan, Zhenyu Tang, **Hongjie Fang**, Chenxi Wang, Cewu Lu#
arXiv preprint, 2025.
22. ChronoFlow Policy: Unifying Past-Future Interaction Flow in Visuomotor Policy Learning
Bokai Lin*, Yifu Xu*, Xinyu Zhan, **Hongjie Fang**, Jialin Tian, Fu-Cheng Zhang, Yong-Lu Li, Cewu Lu, Lixin Yang#
*European Conference on Computer Vision (**ECCV**), 2026.*
23. LIDEA: Human-to-Robot Imitation Learning via Implicit Feature Distillation and Explicit Geometry Alignment
Yifu Xu*, Bokai Lin*, Xinyu Zhan, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.*

24. SIME: Enhancing Policy Self-Improvement with Modal-Level Exploration
Yang Jin*, Jun Lv*, Wenye Yu, **Hongjie Fang**, Yong-Lu Li, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.*

25. Graspness Discovery in Clutters for Fast and Accurate Grasp Detection
Chenxi Wang*, Hao-Shu Fang*, Minghao Gou, **Hongjie Fang**, Jin Gao, Cewu Lu#
*IEEE/CVF International Conference on Computer Vision (**ICCV**), 2021.*

26. Target-Referenced Reactive Grasping for Dynamic Objects
Jirong Liu, Ruo Zhang, Hao-Shu Fang, Minghao Gou, **Hongjie Fang**, Chenxi Wang, Sheng Xu, Hengxu Yan, Cewu Lu#
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.*

Academic Services

Reviewer for journal *RA-L*, *T-CYB*, *T-MECH*, *T-ASE*.
Reviewer for conferences including *ICRA*, *IROS*, *CoRL*, *NeurIPS*, *ICLR*, *CVPR*, *etc*.

Invited Talks

- 03/2024, **Echo AI @ USyd**, *Towards Efficient Robot Imitation Learning from Human Demonstrations.*
- 11/2024, **Zhixingxing**, *Towards Efficient Robot Imitation Learning from Human Demonstrations.*
- 03/2025, **Yang Gao Group @ THU**, *Towards Generalizable Imitation Learning from Human Demonstrations.*
- 08/2025, **3D CVer**, *Towards Generalizable Imitation Learning from Human Demonstrations.*
- 08/2025, **Sharpa**, *Towards Generalizable Imitation via Scalable Data and Robust Policy.*
- 09/2025, **Galbot**, *Building Capable and Generalizable Policies for Robotic Manipulation.*
- 09/2025, **Shenlanxueyuan**, *Towards Generalizable Imitation via Scalable Data and Robust Policy.*
- 10/2025, **The Robot Dexterity Lab @ Duke**, *AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons.*

Teaching

Teaching Assistant, *Algorithm and Complexity*	*Spring, 2022*
Teaching Assistant, *C++ Programming Language (Honor)*	*Fall, 2020*
Teaching Assistant, *Data Structure (Honor)*	*Spring, 2019*
Teaching Assistant, *Linear Algebra (Honor)*	*Fall 2021 & Fall, 2022*
Teaching Assistant, *Mathematical Analysis (Honor)*	*Fall, 2020 & Fall, 2021*